

MOHAMED MAGDY

AI Engineer

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LinkedIn | GitHub | Kaggle

Research Interests

Generative AI, Retrieval-Augmented Generation (RAG & Graph RAG), Physical AI, Computer Vision (Real-Time Object Detection), Digital Twins, Deep Learning Architectures, Robotic Automation Systems, and LLMOps.

Education

October 6 University
B.Sc. Computer Science and Information Systems
Cumulative GPA: 3.37 / 4.00 (Very Good)
Graduation Project: Quality Workflow System (Grade: Excellent)
Military Status: Finally Exempted

2019 – 2023
Egypt

Research & Professional Experience

Global Brands Group (GBG)
AI Engineer

Feb 2026 – Present
Cairo, Egypt

- Graduated as a Top Achiever (Score: 950/950) with the highest commendation in the formal AI Diploma track.
- Architected end-to-end, production-ready pipeline systems utilizing advanced Graph RAG, FastAPI, and Redis caching.
- Designed high-accuracy data retrieval solutions leveraging advanced chunking and Hybrid Search (HNSW & BM25) deployed on Microsoft Azure.
- Utilized LangChain and LangSmith to build, evaluate, and trace LLM-powered autonomous multi-agent systems.

Siemens Industry 4.0 Innovation Center (ITIDA)
AI & Digital Twin Engineer Intern

Jan 2026 – March 2026
New Administrative Capital, Egypt

- Engineered industrial Digital Twin simulations to accurately model, visualize, and optimize factory asset workflows.
- Developed AI-driven predictive maintenance pipelines using real-time mechanical vibration and thermal sensor data.
- Researched spatial optimization algorithms to mitigate manufacturing downtime and lower operational latency.

National Telecommunication Institute (NTI)
Digital Twin Engineering Trainee

Oct 2025 – Jan 2026
Cairo, Egypt

- Completed an intensive 540-hour advanced engineering curriculum, graduating with a final evaluation score of 97%.
- Led the technical system development for the "Adaptive Belt Monitoring" conveyor research project.
- Implemented real-time multi-sensor fusion pipelines and statistical thresholding algorithms for industrial anomaly detection.

ContiLab, Chung-Ang University (CAU)
Visiting Research Fellow (Fully Funded Scholarship)

Mar 2025
Seoul, South Korea

- Selected for a highly competitive, fully funded international research scholarship focused on Construction Safety Management.
- Collaborated with graduate researchers to investigate state-of-the-art computer vision systems in real-world hazard environments.
- Explored the theoretical integration of real-time object detection models and Unity-based Digital Twins for predictive site safety.

Digital Egypt Pioneers (DEPI) Program MCIT
AI & Data Science Trainee (IBM Track)

Apr 2024 – Oct 2024
Cairo, Egypt

- Mastered foundational deep learning frameworks, advanced statistical modeling, and relational database systems.
- Engineered a full-stack machine learning capstone project optimized for business decision analytics.

Fawry MSME Finance
Data Analysis Intern

Jul 2024 – Sep 2024
Cairo, Egypt

- Extracted and preprocessed complex transaction datasets using SQL and Python (Pandas) to analyze regional credit risk patterns.
- Designed executive-level interactive KPI dashboards in Power BI to optimize cross-departmental marketing strategies.

Zewail City of Science and Technology

Aug 2023 – Nov 2023

AI Intern

Cairo, Egypt

- Completed rigorous coursework covering core machine learning mathematical models, optimization, and feature engineering.
- Applied exploratory data analysis (EDA) techniques to design robust data preprocessing pipelines.

Academic & Research Projects

Tomato Leaf Disease Detection Pipeline

Feb 2026 – May 2026

Deep Learning & Computer Vision Framework

Cairo, Egypt

- Developed an end-to-end, high-precision deep learning computer vision pipeline targeting >96% classification accuracy.
- Implemented an optimized single-stage YOLO architecture specialized in joint localization and multi-class disease classification.
- Engineered advanced data augmentation and preprocessing routines to maintain inference stability under highly variable illumination.

Adaptive Belt Monitoring System (ABM)

Oct 2025 – Jan 2026

Industrial Digital Twin Core Project (NTI Capstone)

Cairo, Egypt

- Designed a real-time Cyber-Physical Digital Twin system utilizing structural vibration data to classify mechanical tension states.
- Built an integrated asynchronous networking bridge connecting Python-based inference models with Unity 3D via the MQTT protocol.
- Trained highly accurate multi-class classification models to predict imminent mechanical failure types before onset.

Quality Workflow System (Graduation Project)

Aug 2023

Process Digitization & Architecture Optimization

Grade: Excellent

- Architected a scalable full-stack web application to digitize and automate intricate paper-based workflow logic.
- Presented project findings on workflow performance optimization methodologies at the 16th Undergraduate Research Forum (UGRF).

Technical Skills

Generative AI & LLMOps: LLMs (GPT, LLaMA), Graph RAG, Standard RAG, LangChain, LangSmith, Prompt Engineering, Vector Databases

ML Frameworks & Libraries: PyTorch, TensorFlow, Scikit-Learn, Keras, Pandas, NumPy

Computer Vision & Architectures: CNNs, Transformers, YOLO, Segment Anything Model (SAM), Real-time Object Detection

Cloud, Backend & Tools: Microsoft Azure, Azure OpenAI, FastAPI, Redis Caching, Docker, Git, Unity (Digital Twins), MQTT, n8n

Data & Databases: SQL, Power BI, Tableau, Advanced Excel

Languages: Arabic (Native), English (Fluent, Official IELTS Score: 6.5)

Certifications

- **Certificate of Achievement - AI Diploma** (Top Achiever, 950/950) (GBG Academy)
- **IBM Data Science Professional Certificate** (IBM)
- **Linear Algebra for ML and Data Science** (DeepLearning.AI)
- **HCIA-Big Data Certification** (Huawei)
- **Artificial Intelligence Training** (IMPACT)
- **C++ Fundamentals** (Coach Academy)
- **Digital Twin Engineering Track** (97% Score) (NTI)
- **Supervised Machine Learning** (DeepLearning.AI)
- **Getting Started with Deep Learning** (NVIDIA)
- **Intro to AI & Applications** (Zewail City)
- **Mini ITI Diploma** (ITI)